THE NOVA RESEARCH CURRICULUM

Physics

Grade 7 · Term 3

Student edition

Outcome: PH-7.4 gravity, mass and weight, orbits, tides and seasons.

Lesson notes

- **Gravity, mass and weight (Weeks 1–4):** gravity pulls every mass towards every other mass. Mass (kg) is the amount of matter and does not change; weight (N) is the pull of gravity on it. Weight = mass × g; on Earth g ≈ 10 N/kg; on the Moon about one sixth of that. Practical: weigh objects with a newton meter.
- **Orbits and tides (Weeks 5–8):** gravity keeps the Moon and satellites in orbit round the Earth and the planets round the Sun. Tides: the Moon's gravity pulls the oceans; most coasts have two high tides a day. Karachi's coast as a local example.

  ▶ **Try it on NovaSchool:** Tides: the Moon pulls the oceans — open **Simulations** on your dashboard.
- **Seasons (Weeks 9–12):** the Earth's axis is tilted (about 23.5°). When our half of the Earth leans towards the Sun, days are longer and sunlight is more direct: summer. Seasons are not caused by distance from the Sun.

  ▶ **Try it on NovaSchool:** Seasons: the tilted Earth — open **Simulations** on your dashboard.

Workbook pages

Page	Task
1	Weight on Earth and the Moon for six masses.
2	Diagram of the Moon's orbit and tides.
3	Seasons diagram with explanation.

Question bank

#	Outcome	Question	Your answer
1	PH-7.4	Weight is measured in ...	kilograms / newtons / metres / litres
2	PH-7.4	A 50 kg student's weight on Earth (g = 10 N/kg)?	50 N / 5 N / 500 N / 5,000 N
3	PH-7.4	On the Moon your mass ...	becomes six times smaller / stays the same / doubles / becomes zero
4	PH-7.4	Tides are mainly caused by the ...	wind / Sun's heat / Moon's gravity / Earth's core

#	Outcome	Question	Your answer
5	PH-7.4	Seasons happen because …	the Earth moves closer to the Sun / the Earth's axis is tilted / the Moon blocks the Sun / clouds change
6	PH-7.4	Gravity keeps satellites in orbit. True or false.	True / False
7	PH-7.4	Explain the difference between mass and weight.	_____